

SHREYAS KUMAR BHAKTA

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New York City . Open to Relocate

Professional Summary

Backend Software Engineer with 5+ years of experience building production-grade distributed systems across fintech and mobility domains. Strong expertise in Java, Spring Boot, Kafka, Redis, microservices, AWS, Docker, and Kubernetes. Proven impact includes reducing API latency by 40%, enabling payment workflows for major banking clients, and raising test coverage to 80%+. Hands-on experience with LangChain, RAG pipelines, and vector databases. Recent M.Eng graduate from Stevens Institute of Technology (4.0 GPA).

Technical Skills

Languages: Java, Python, JavaScript, TypeScript, Kotlin, C++, SQL, Bash

Backend: Spring Boot, Node.js, Express.js, REST API Design, Microservices, Event-Driven Systems, JWT, OAuth 2.0, gRPC basics

Databases and Caching: PostgreSQL, MongoDB, MySQL, Firebase, Redis, NoSQL, Data Modeling, ETL Pipelines

Messaging and Streaming: Apache Kafka, RabbitMQ basics, Message Queues, Event Streaming

Cloud and DevOps: AWS EC2, AWS S3, AWS Lambda, GCP basics, Docker, Kubernetes, Jenkins, CI/CD, SonarQube, Prometheus, Grafana

AI and GenAI: LangChain, LlamaIndex, RAG Pipelines, Vector Databases, Pinecone, FAISS, TensorFlow.js, OpenAI API, Hugging Face, Prompt Engineering

Testing and Quality: JUnit, Jest, Mockito, Postman, Swagger, TDD, Code Reviews, RAGAS Evaluation

Security: OWASP Guidelines, DevSecOps, SSL/TLS, Vulnerability Scanning, Penetration Testing, Metasploit, Wireshark

Practices: Agile, Scrum, System Design, SDLC, Git, Jira, Confluence, Technical Mentorship

Professional Experience

Member of Technical Staff - Software Engineer

Sep 2023 – Jan 2025

Vymo - Fintech SaaS, Enterprise Sales Intelligence Platform

Bengaluru, India

- Reduced end-to-end API latency by 40% by profiling Java/Spring Boot microservices, removing redundant calls, optimizing database queries, and applying request-level caching.
- Architected and integrated a payment aggregator module for HDFC Bank, enabling real-time payment processing and reconciliation while unlocking 90% of collection-related platform features.
- Led a cross-functional team of 5 engineers to design and deliver a settlement module in collaboration with HDFC Bank stakeholders, on schedule through Agile/Scrum execution.
- Improved observability by integrating Prometheus-based monitoring and alerting, reducing mean time to detect production incidents by 30%.
- Conducted structured code reviews, enforcing engineering standards and reducing post-release defects.

Software Development Engineer

Mar 2022 – Jul 2023

SIXT Research and Development - Global Mobility and Car Rental Technology

Bengaluru, India

- Increased Digital Checkout Service adoption from 2% to 6%, supporting 6,000+ daily checkouts by developing configurable REST APIs in a microservices architecture.
- Reduced API response time to under 300 ms by implementing Redis-based request-level caching, reducing downstream service load.
- Raised unit and integration test coverage to 80% using JUnit and Mockito; integrated SonarQube into CI/CD pipelines for continuous quality enforcement.
- Reduced service-level latency by 30–40% and improved event traceability by implementing Apache Kafka funnels for asynchronous tracking and validation workflows.

Platform Security and Development Engineer

Jul 2021 – Dec 2021

Brane Enterprises Pvt. Ltd. - Enterprise Technology Solutions

Hyderabad, India

- Identified 11 security vulnerabilities through penetration testing and automated scanning using Metasploit and Wireshark, reducing platform attack surface by 50–60%.
- Implemented DevSecOps practices by integrating SAST and DAST scans into CI/CD pipelines and applying OWASP guidelines across the SDLC.

Android App Development Intern

Mar 2021 – May 2021

JSW Pvt. Ltd. - Steel and Infrastructure Conglomerate

Bellary, India

- Developed a Kotlin-based internal Android application for employee service access, reducing access latency by 70% over the legacy process.

Android App Developer Schaffen Softwares

Aug 2020 – Feb 2021
India

- Built a Kotlin utility app with QR code generation, QR scanning, and text translation; promoted within 3 months for delivery quality and UI contributions.

Projects

MindMesh - Cross-Platform Wellness App Node.js, LangChain, LangGraph, FAISS, MongoDB, Firebase, Redis

- Built an agentic AI insight engine using LangChain and LangGraph that autonomously decides whether to retrieve from user health history, query a wellness knowledge base, or escalate to a more capable model based on query complexity, reducing unnecessary API calls by 35%.
- Designed a hybrid retrieval pipeline combining dense vector search with recency-weighted scoring; the agent applies tool-use patterns with a human-in-the-loop approval gate before surfacing health recommendations, ensuring responsible agentic deployment.
- Validated agentic output quality using RAGAS metrics including faithfulness, answer relevancy, and context precision; implemented secure REST APIs with JWT, Redis caching, and offline-first Firebase sync supporting 1,000+ concurrent users.

SkyChoice - Airline Decision Support System

React, TypeScript, Node.js, Express.js

- Built an AI-assisted decision support system where an orchestration agent selects and runs the optimal scoring algorithm from six multi-criteria methods including TOPSIS, MAUT, VIKOR, and PROMETHEE II based on traveler profile and route context.
- Integrated Monte Carlo simulation with 500 iterations for probabilistic sensitivity analysis across 7 criteria, with the agent dynamically adjusting recommendation confidence based on data availability and simulated uncertainty.
- Connected live flight pricing via SerpAPI Google Flights with automatic fallback to BTS 2024 historical baseline, ensuring agentic pipeline resilience under external API failure.

Ducktor - Real-Time Posture Detection Desktop App

Electron, TensorFlow.js, MoveNet, JavaScript

- Built a locally-running AI agent using TensorFlow.js with MoveNet that continuously monitors posture via webcam, classifies deviations from 17 skeletal keypoints, and autonomously triggers corrective alerts with sub-second inference latency and no cloud dependency.
- Shipped a working agentic MVP within 24 hours at Quack Hacks, implementing the full perception-decision-action loop from pose detection through configurable alert thresholds to user notification.

Railway Route Optimization

Flutter, Dart, Data Structures and Algorithms

- Engineered a Flutter application using Dijkstra's algorithm to calculate shortest railway routes, rendered on an interactive map interface.

Education

Stevens Institute of Technology

Jan 2025 – May 2026

Master of Engineering in Engineering Management, GPA: 4.0

Hoboken, NJ, USA

Coursework: System Design, Product Management, Engineering Economics, Agile Project Leadership, Decision Analysis

Dayananda Sagar College of Engineering

Aug 2018 – Jun 2022

Bachelor of Engineering in Information Science and Engineering

Bengaluru, India

Coursework: Data Structures and Algorithms, DBMS, Operating Systems, Computer Networks, Machine Learning, Software Engineering